

Progress Report: Hybrid Rover Path Planning and Reflex Module

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1 Overview

This progress report summarizes the development of the **Hybrid Rover Path Planning and Reflex Module** implemented as part of the planetary exploration simulation. The goal of this module is to enable an autonomous rover to efficiently navigate toward a target sample zone while avoiding hazards and maintaining sufficient battery charge. The system combines a *reflex-based decision layer* for immediate response and an *A*-based planning layer* for long-term optimal path computation.

2 Implementation of the Reflex Module

The reflex module is responsible for real-time decision-making under dynamic and uncertain terrain conditions. It ensures that the rover reacts instantly to local threats or resource shortages before invoking the global planner.

The three reflex rules are prioritized in the following order:

1. **Recharge Reflex:** When the rover's battery level falls below a safety threshold (25%), it halts movement and attempts to recharge. This prevents premature mission failure due to energy depletion.
2. **Radiation Avoidance Reflex:** If the rover detects that its next move leads to a high-radiation cell, it immediately backtracks and requests a new path from the planner. This rule maintains rover integrity under environmental hazards.
3. **Halt Reflex:** Upon reaching the `SAMPLE_ZONE`, the rover terminates all activity and records a successful mission. This reflex ensures correct mission termination and state recording.

The reflex module thus acts as a **safety net**—prioritizing survival and mission continuity over speed or distance optimization.

3 Implementation of the Path Planning Module

The path planning component utilizes the *A* search algorithm*, a well-established heuristic-driven technique for graph-based route optimization. The planner continuously computes the most efficient path from the rover's current position to the goal, using both actual traversal cost and heuristic estimates.

3.1 Cost and Heuristic Functions

- The **cost function** $g(n)$ measures the battery energy consumed when moving across various terrain types (Flat, Rough, Hilly, etc.).
- The **heuristic function** $h(n)$ estimates the remaining distance to the goal and guides the planner's search.

Four heuristic functions were implemented and analyzed:

1. **Manhattan Distance**: Calculates the total grid distance between the current and goal cells without diagonal movement. It is admissible but conservative.
2. **Euclidean Distance**: Measures the straight-line distance. It is more natural for open terrains and provides smoother guidance.
3. **Chebyshev Distance**: Allows diagonal moves by considering the maximum of coordinate differences. It balances path length and flexibility.
4. **Weighted Euclidean ($w=1.1$)**: A modified, inadmissible version of the Euclidean distance. It slightly overestimates the cost to encourage faster, goal-directed search at the expense of strict optimality.

4 Descriptive Comparison of Heuristics

The comparative analysis (detailed quantitatively in a separate Jupyter notebook) revealed key behavioral differences between the four heuristics. This section qualitatively summarizes those findings.

4.1 Search Efficiency

The **Weighted Euclidean heuristic** consistently produced shorter computation times and fewer node expansions due to its aggressive goal bias. The **Euclidean** and **Chebyshev** heuristics achieved balanced exploration, while the **Manhattan** heuristic was slower and more exhaustive because of its restricted movement model.

4.2 Path Optimality

In terms of energy consumption, **Weighted Euclidean** achieved near-optimal battery usage by avoiding redundant exploration. However, it occasionally produced slightly suboptimal paths due to its inadmissible nature. **Euclidean** and **Chebyshev** remained robust and consistent in path quality, while **Manhattan** often yielded longer, less efficient routes.

4.3 Reliability and Stability

The **Chebyshev** and **Weighted Euclidean** heuristics exhibited the highest mission completion rates across multiple epochs. Their ability to combine flexibility with focused search proved advantageous in dynamic environments. In contrast, the **Manhattan** heuristic frequently failed to converge within the step limit, and the **Euclidean** heuristic occasionally led to stalled progress in cluttered terrains.

5 Conclusions

The hybrid framework successfully integrates a reactive reflex layer and a strategic planning layer. While all four heuristics achieved reasonable performance, the **Weighted Euclidean** heuristic provided the best trade-off between computational efficiency, reliability, and energy efficiency. It demonstrated that slight heuristic relaxation can enhance performance in constrained, dynamic systems like planetary exploration rovers.

Future work will explore adaptive heuristic weighting and real-time learning mechanisms to further improve the rover's autonomy and responsiveness.